

# Daily Quiz - 2 (ELL782/ELL7282) - Computer Architectures

## Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.
- For Boolean expressions, X' denotes NOT X, + denotes OR, and adjacent variables denote AND.

Points: 10/10

1. Name \* 

test

2. Entry Number \* 

test

✓ **Correct** 1/1 Points

3. 

Let  $F = A'B + AC + BC$ . Which of the following is equal to  $F'$ ?

☐  $(A' + B)(A + C)$

☐  $(A + B)(A' + C')$

☒  $(A + B')(A' + C')$

☐  $AB' + AC'$

✓ **Correct** 1/1 Points

4. Select all that apply. 

Let  $X = (110101101011)_2$ . Which representations are equal to  $X$ ?

☒  $0xD6B$

☒  $(6553)_8$

☒  $(3435)_{10}$

☐  $0x6D3$

✓ **Correct** 1/1 Points

5. The 5-bit pattern 11010 is interpreted, in order, as sign-magnitude, one's complement, two's complement, and bias-15. Which tuple of values is correct?



☐  $(-5, -10, -6, +11)$

☒  $(-10, -5, -6, +11)$

☐  $(-10, -6, -5, +11)$

☐  $(-10, -5, -6, -11)$

✓ **Correct** 1/1 Points

6. In a 5-bit bias-15 representation,  $F(u) = 10011$  and  $F(v) = 01100$ . Which bit pattern represents  $u + v$ ? 

☐ 11111

☐ 00001☐ 01111☒ 10000

✓ **Correct** 1/1 Points

7. In 6-bit two's-complement arithmetic, what is the stored result of  $011001 - 101101$  and does signed overflow occur? 

☐ 001100; no signed overflow☒ 101100; signed overflow occurs☐ 101100; no signed overflow☐ 001100; signed overflow occurs

✓ **Correct** 1/1 Points

8. A 6-bit two's-complement register contains 100000. Its two's complement is computed in the same 6 bits by inverting all bits and adding 1. What is obtained? 

☒ 100000; it still denotes -32 because +32 is not representable☐ 000000; it denotes zero☐ 011111; it denotes +31☐ 100001; it denotes -31

✓ **Correct** 1/1 Points

9. A 5-bit two's-complement value  $x = 10110$  is sign-extended to 8 bits, and then 00011101 is added. What is the stored 8-bit result? 

- ☐ 00010011; signed overflow occurs
- ☒ 00010011; no signed overflow
- ☐ 00110011; no signed overflow
- ☐ 11110011; no signed overflow

✓ **Correct** 1/1 Points

10. Which of the following 5-bit two's-complement additions cause signed overflow? 

Select all that apply.

- ☒ 01100 + 01011
- ☒ 10000 + 11111
- ☒ 10010 + 10101
- ☐ 01111 + 10001

✓ **Correct** 1/1 Points

11. In an 8-bit two's-complement system, the register pattern 00101101 represents  $x$ . Let  $p$  encode  $-x$  and  $q$  encode  $-x - 1$ . Which ordered pair  $(p, q)$  is correct? 

- ☒ (11010011, 11010010)
- ☐ (11010010, 11010011)
- ☐ (00101101, 11010010)
- ☐ (11010011, 00101100)

✓ **Correct** 1/1 Points

12. An 8-bit adder produces 10000000 from  $01111111 + 00000001$ . Which interpretation is correct? 

- ☐ Unsigned overflow occurs. Two's complement: no overflow; the result is -128.
- ☒ Unsigned: 128, no overflow. Two's complement: signed overflow; the stored pattern denotes -128.
- ☐ Neither interpretation overflows; the result is +128 in both interpretations.
- ☐ Both interpretations overflow because the most significant bit becomes 1.

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